Data Quality Documentation

Overview

The data quality assessment was carried out on the FinTrust Digital Bank dataset to check its completeness, consistency, validity and reliability before performing SQL analysis, exploratory data analysis and dashboard development.

The project used two datasets: Customers and Transactions. The Customers dataset contained 1,500 records and 12 columns, while the Transactions dataset contained 12,000 records and 11 columns.

Data Quality Checks

1. Completeness

- The Customers dataset had no missing values.
- The Transactions dataset had 96 missing values in Device_Type and 96 in Location.
- The missing values were considered during analysis to avoid misleading conclusions.

2. Uniqueness

- Customer IDs were unique in the Customers dataset.
- No duplicate transaction records were identified during the initial checks.
- Transaction IDs were checked to support accurate transaction counting.

3. Validity

- Customer ages ranged from 18 to 65.
- Engagement scores fell within the expected range of 10 to 100.
- Transaction amounts were non-negative, with a minimum of NGN 100.20.
- Transaction dates were converted to datetime format in Python for time-based analysis.

4. Referential Integrity

- All customer IDs recorded in the Transactions dataset matched customer IDs in the Customers dataset.
- No unmatched customer records were identified.
- A relationship between the Customers and Transactions tables was used to support joined analysis and Power BI reporting.

5. Outlier Detection

- The IQR method was used to identify unusually high transaction amounts.

- The upper outlier threshold was approximately NGN 117,922.86.
- A total of 1,474 transactions were flagged as potential outliers.
- These transactions were retained for investigation because high amounts do not automatically indicate data errors or fraud.

6. Consistency

- Transaction types, channels, statuses and customer segments were reviewed during the analysis.
- Transaction counts and values were compared across SQL, Python and Power BI where applicable.
- KPI calculations were reviewed to ensure percentages used the correct denominators and filter context.

Data Quality Limitations

- Missing device and location information limited some channel and location-based analysis.
- Potential outliers required further investigation before being classified as errors.
- Risk review flags indicated transactions requiring review, not confirmed fraudulent activity.
- The dataset was synthetic, so findings represented the supplied data rather than actual bank performance.

Conclusion

The data quality checks identified missing values in two transaction columns and several potential high-value outliers. Customer references were consistent across both datasets, and no duplicate records were identified in the initial assessment. The data was suitable for the planned analysis, subject to the documented limitations and final validation of dashboard measures.